

The empirical recurrence for OEIS A263220: recovering a conjecture that is not written down

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Abstract

OEIS A263220 records “Empirical recurrence of order 89”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 375$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 89, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 89 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A263220 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with each row and column divisible by 5, read as a base-3 number with top and left being the most significant digits.”. It has offset 1 and begins

7, 343, 12601, 475539, 22342751, 1094138875, 53375395657, 2607171566653, ...

The entry states:

Empirical recurrence of order 89 (see link above)

As of the “Last modified” line on the live entry (Oct 12 2015 11:47:07, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 625$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 375$ of the 625, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 375$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 750$ exact terms of the model returns order 89 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{89} c_i a(n-i),$$

c_1	49	c_{24}	−7246391174398181959117848	c_{47}	
c_2	0	c_{25}	355073167545510915996774552	c_{48}	−39055964
c_3	0	c_{26}	0	c_{49}	1913742270
c_4	117537	c_{27}	0	c_{50}	
c_5	−5759313	c_{28}	16110282251562979382678116104	c_{51}	
c_6	0	c_{29}	−789403830326585989751227689096	c_{52}	1014412282
c_7	0	c_{30}	0	c_{53}	−4970620183
c_8	−3657133446	c_{31}	0	c_{54}	
c_9	179199538854	c_{32}	−24708812441451346963168924757970	c_{55}	
c_{10}	0	c_{33}	1210731809631116001195277313140530	c_{56}	−1843696747
c_{11}	0	c_{34}	0	c_{57}	90341140633
c_{12}	53128681405190	c_{35}	0	c_{58}	
c_{13}	−2603305388854310	c_{36}	26470546327756866922621671627202770	c_{59}	
c_{14}	0	c_{37}	−1297056770060086479208461909732935730	c_{60}	233950154147
c_{15}	0	c_{38}	0	c_{61}	−114635575532
c_{16}	−435053443034280765	c_{39}	0	c_{62}	
c_{17}	21317618708679757485	c_{40}	−19914959221727706906791121040771738596	c_{63}	
c_{18}	0	c_{41}	975833001864657638432764930997815191204	c_{64}	−205655915335
c_{19}	0	c_{42}	0	c_{65}	10077139851427
c_{20}	2199539695828751312109	c_{43}	0	c_{66}	
c_{21}	−107777445095608814293341	c_{44}	10529515064229085947116719852142171675172	c_{67}	
c_{22}	0	c_{45}	−515946238147225211408719272754966412083428	c_{68}	12308177526527
c_{23}	0	c_{46}	0	c_{69}	−60310069879983

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 89, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $89 + 4$ terms removed returns order 89 as well, so $\deg q_{\min} = 89 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $89 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 89 that this sequence satisfies — and that family turns out to have exactly one member.

References

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